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The *Cool Teens* CD-ROM for anxiety disorders in adolescents

A pilot case series

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Abstract Five adolescents received a multimedia CD-ROM containing a self-help treatment program for young people with an anxiety disorder. Participants used the 8-module *Cool Teens* CD-ROM over a 12-week period on a home computer. Every 2 weeks, they received a brief telephone call from a clinical psychologist to monitor symptoms and progress and to discuss any problems with understanding content or implementing techniques. Based on structured interviews, two participants (40%) no longer met diagnostic criteria (self-report ADIS) for at least one clinical anxiety disorder immediately following

treatment and these same participants no longer met diagnostic criteria for any clinical anxiety disorder at 3-month follow-up. Two other participants failed to make gains based on diagnostic criteria, but showed improvement in anxiety symptoms for one main fear. Participants were generally satisfied with the multimedia content, the modules, and the delivery format of the program.

Key words anxiety disorders – adolescents – computer-based CBT – multimedia – self-help

Introduction

Despite clinical anxiety being one of the most common mental health conditions in adolescents [5] and that many efficacious cognitive behavioural therapy (CBT) programs have been developed (reviews, [3, 10]), only one in four young people receive professional help for this problem [18]. Identified barriers with traditional treatments and services include confidentiality, stigma, cost, geographic or social isolation, content appropriateness, format of engagement, and waiting time [2, 28].

In their review of 13 studies involving 498 adolescents and children, James et al. [10] reported that CBT was effective in 56% of cases and that the effects were durable. Individual, group, or parental/family

formats all produced similar outcomes. However, older adolescents were under-represented in the review and clinical experience suggests that anxiety in this group presents additional treatment challenges due to difficulties with engagement and motivation at this developmental stage.

More “creative and innovative” approaches to treatment are worth exploring [9], and one promising format is computer-based treatment [1, 4, 12, 19]. Advantages include cost effectiveness (reduced therapist time), time and travel convenience, engaging content, potential to reach those reluctant to make face-to-face contact, and broader service dissemination, particularly to rural participants. Disadvantages include extensive time and resources needed to develop such programs, difficulty monitoring program use, potential technical problems, and lack of

suitability for some clients. There are no existing published studies on computer-based therapy programs for adolescents with anxiety.

A multidisciplinary team at Macquarie University's Centre for Emotional Health has recently produced the *Cool Teens CD-ROM*, a home-based therapy option for anxious adolescents [6, 8], allowing individuals to move at a pace and intensity to suit their specific problems. Content focuses on social phobia (SP) and generalized anxiety disorder (GAD), but also covers separation anxiety disorder (SAD), specific phobias (SpP), and obsessive-compulsive disorder (OCD). Feedback to a prototype rated the interactive format of this program positively, and many adolescents reported it acceptable as a therapy option [7]. This paper presents a case series on early users of the *Cool Teens CD-ROM*.

Methods

■ Participants

Four females (aged 14–16 years) and 1 male (aged 16 years) were recruited from successive parental referrals to the Macquarie University Anxiety Research Unit (MUARU), Sydney, Australia between March and May 2006. Inclusion criteria were: a principal diagnosis of an anxiety disorder, age between 14 and 18 years, and access to a home computer. Exclusion criteria included problems requiring immediate attention (e.g., suicidal ideation, self-harm, abuse, psychosis) and significant learning difficulties. Participants were required to stabilise current medication and to not engage in other active therapy during the study period. Four participants had a principal diagnosis of GAD and one had SAD. Many had additional anxiety diagnoses. The research was approved by the Macquarie University Ethics Review Committee and each adolescent and a parent gave written consent.

■ Measures

Pre- and post-treatment diagnoses and outcomes were clinically assessed using the Anxiety Disorders Interview Schedule for Children (ADIS-C-IV) [24]. This provides a DSM-IV diagnosis and is reliable for child and adolescent anxiety disorders ($\kappa = 0.8\text{--}1.0$) [15]. ADIS-IV ratings are reported as a clinician severity rating (CSR) on a 9-point scale (0 = not at all disturbing/disabling; 8 = very severely disturbing/disabling; a rating under 4 is sub-clinical), based on adolescent report, parent report, and therapist's combined interpretation. The structured interview

was conducted face-to-face for participants in the Sydney area and over the telephone for participants outside the Sydney area. All post-treatment ADIS assessments were conducted over the telephone. Previous investigations have demonstrated that telephone administration has high validity [13].

Pre- and post-treatment anxiety symptoms were measured using the Spence Children's Anxiety Scale (SCAS) [26, 27]. This 38-item measure assesses both adolescent (SCAS-C) and parent (SCAS-P) reports of symptoms, closely corresponding to those in the DSM-IV. It has good psychometric properties with respect to reliability and validity. Negative beliefs were assessed using the Children's Automatic Thoughts Scale (CATS) [22], a 40-item self-report measure designed to assess negative self-statements. This scale has been shown to have good discriminant validity for each of the subscales across childhood disorders, including anxiety, and good convergent validity with related anxiety scales [21, 23]. During a therapist contact telephone call every 2 weeks, a shortened version of the SCAS-C and an Anxiety Symptom Tracking Measure designed to assess each participant's self-identified main fear were administered. Participant satisfaction was measured using a 10-question adapted Barriers to Treatment Participation Scale [11] and a Preferences and Attitudes Questionnaire.

■ Treatment

The *Cool Teens CD-ROM* is a multimedia program incorporating text, graphics, video, voiceover, and graphics. It comprises 8 modules covering the following therapeutic components: Module 1—Psychoeducation, Module 2—Goal Setting, Modules 3 and 5—Cognitive Re-structuring, Modules 4 and 6—Exposure, Module 7—Coping Skills, and Module 8—Maintenance. Each module takes 15–30 min to complete and contains information, interactive exercises, hypothetical scenarios, case studies, and practice tasks [6].

The content of the CD-ROM modules is based on material from the well-established *Cool Kids* anxiety program that delivers these same cognitive behavioural skills through 10-week therapist-guided group therapy [16]. This program has been shown to be efficacious in clinical settings, school settings, telephone based delivery, and bibliotherapy format [14, 17, 20].

In the current study, use of the CD-ROM was supported by some therapist contact. A clinical psychologist made a telephone call to each participant every 2 weeks to ask a set of questions regarding progress and understanding of material. The purposes of the calls were to provide motivation, keep participants

moving through modules at a good pace, and clarify any issues regarding the content. Minimal therapeutic assistance was included if requested. The adolescent was reminded of the clinic's contact details if they need help with strong negative feelings.

Procedures

Potential participants were screened briefly to ascertain suitability for study inclusion and were provided with information and consent forms. To assess the diagnostic status of participants, ADIS-IV interviews were conducted by a clinical psychologist. Pre-treatment questionnaires (SCAS-C, SCAS-P, & CATS) were sent by mail for completion and a pre-paid envelope was provided for their return.

Once questionnaires and consent forms were received, participants were sent the *Cool Teens* CD-ROM for use over an initial period of 12 weeks. They were asked to work through the self-help program and were informed they would receive a brief call from a psychologist every 2 weeks to assess progress. They were told they would be re-assessed after 12 weeks and should aim to complete all of the modules by then. Participants were provided with contact details for the clinic to give them a sense of connection and to provide access to professional help in case of an emergency.

After 12 weeks of therapy, each participant completed a post-treatment telephone ADIS interview and returned Barriers to Treatment Scale and User Preferences and Attitudes questionnaires. Each participant received \$30.00 to compensate for time spent on the research. Participants were encouraged to continue using the CD-ROM for a further 12 weeks, and were told that the same re-assessment and payment procedure would be carried out after that follow-up period.

Results

Table 1 shows the pre- and post-treatment and 3-month follow-up scores for the clinical ADIS interview. Four participants completed at least six of the eight modules within the 12 weeks. The length of the scheduled telephone support calls ranged from 5–20 min per session (mean = 13.5, SD = 6.82).

Participant 1, a 15-year old female with GAD and 3 co-morbid anxieties, completed all 8 modules within the initial treatment period and reported regular consistent use of the program. Her most positive gain was a large improvement in anxiety rating for her main fear as measured by the Anxiety Symptom Tracking Measure. Her parent SCAS score demonstrated some reduction in anxiety but no changes in ADIS diagnosis or severity were observed.

Participant 2, a 16-year old female with GAD and a co-morbid SpP, completed 6 modules. At post-treatment she improved greatly on the self-reported anxiety rating for her main fear. While her ADIS-C rating for GAD was reduced from 6 to 4, this improvement was not supported by ADIS-P and self-report measures data. In addition, she now reported social anxiety fears that met criteria for SP, despite having denied their presence at pre-assessment. At follow-up, her ADIS-C rating was slightly reduced from the pre-treatment level, and some gains were made in her self-reported SCAS and CATS scores.

Participant 3, a 14-year old female with SAD along with secondary GAD and a SpP, completed 7 modules. Post-treatment, her ADIS rating for SAD showed slight improvement and she reported that she was “just getting to” the major steps in her exposure hierarchy. Severity ratings for her secondary diagnoses had both reduced to subclinical levels based on combined ADIS data. At follow-up, her ADIS-C rating for SAD was one and the ADIS-P and combined ratings for her three anxiety disorders were at subclinical

Table 1 Pre, post, and follow-up ADIS severity ratings for their primary diagnosis only

Measure		Case 1 F, 15, GAD	Case 2 F, 16, GAD	Case 3 F, 14, SAD	Case 4 F, 15, GAD	Case 5 M, 16, GAD
ADIS (0–8) (adolescent only report)	Pre	6	6	6	6	6
	Post	6	4	5	2	2
	Follow up	6	5	1	–	0
ADIS (0–8) (parent only report)	Pre	7	6	6	6	6
	Post	6	7	6	2	–
	Follow up	7	7	3	–	6
ADIS (0–8) (adolescent & parent combined report)	Pre	7	6	6	6	6
	Post	6	7	5	2	2
	Follow up	7	7	3	–	5
Total # Anxiety Diagnoses (combined report)	Pre	5 (2 SpP)	2	3	5 (4 SpP)	1
	Post	5 (2 SpP)	3	1	1	0
	Follow up	6 (3 SpP)	3	0	–	0

ADIS-C anxiety disorders interview schedule for Children (DSM-IV); ADIS-P ADIS-parent version; ADIS combined clinician rating from child and parent reports

levels. Her parental SCAS reports were much improved.

Participant 4, a 15-year old female with GAD and four co-morbid SpPs, completed two modules before discontinuing treatment after week 4 because she reported she no longer had anxiety and required no further therapy. This participant and her mother both reported that the early CD content, especially the psychoeducation module, had helped her discuss and work on her problems more with her family. Her post-treatment ratings (ADIS-C and ADIS-P) were reduced to subclinical levels for GAD and all but one SpP.

Participant 5, a 16-year old male with GAD, completed all 8 modules and showed an improvement to a subclinical level as shown by an ADIS-C anxiety rating reduction from 6 to 2. This finding was supported by some gains in SCAS and CATS scores and by the paternal SCAS-P scores. At follow-up, his ADIS-C anxiety rating remained subclinical and was even further reduced (to 0).

All program modules and media components were rated positively. Some barriers were identified, with "finding time" rated as the greatest difficulty. Specific likes and dislikes were reported and suggestions for improvements were made, but seemed to reflect personal preferences rather than recurring program weaknesses.

Discussion

The pilot research reported here was a case series that evaluated several clinical outcomes with five adolescents who used the *Cool Teens CD-ROM* for the treatment of diagnosed anxiety disorders over a 12-week therapy period with a 3-month follow-up. Post-treatment assessment showed 2 participants (40%) had anxiety severity ratings that were reduced to a subclinical level for at least 1 clinical anxiety

disorder. At 3-month follow-up, these two participants no longer met diagnostic criteria for any clinical anxiety disorder. All of the adolescents were generally satisfied with the multimedia content, modules, and delivery format of the program.

The adolescent who had the greatest overall improvement was the only one who had no secondary diagnoses. The participant who had the next best overall gains had two secondary diagnoses but her primary problem was SAD rather than a broader disorder. In contrast, the participants who had a larger number of secondary diagnoses and more complex clinical presentations made the least overall improvements. This is consistent with a previous study showing that higher levels of pre-treatment symptoms predicted poorer post-treatment outcomes [25].

Not all measures indicated improvements. Indeed, participant scores on some scales increased from pre- to post-treatment. One possibility is that the program increased adolescents' awareness of their anxieties. It is important to note that only two of the five participants completed all eight modules within the 12-week time-frame, mainly due to not being able to "find time".

The overall level of effectiveness from this pilot evaluation of the *Cool Teens CD-ROM*, with its mixed findings for various participants, may suggest two broad possibilities. First, there may be ways to improve the program by enhancing the CD-ROM's various components with the goal of increasing user compliance, motivation, and benefit. Second, this self-help format may never be expected to do as well as therapist-led treatment in terms of clinical efficacy. Based on the positive user feedback data, this format also has other advantages such as increased accessibility and acceptability for this rather difficult to engage population.

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